

DAV-2 Problem Solving 3rd session

Class starts at 9:10pm

Mean weight of 4 students = 40 kg
when teacher includes (total 5 people), mean becomes = 42 kg

$$\begin{aligned}\text{Total wt of 4 students} &= 4 \times 40 \\ &= 160 \text{ kg}\end{aligned}$$

$$5 \text{ people} = 42 \times 5 = 210 \text{ kg}$$

$$\text{Teacher's wt} = 210 - 160 = \underline{\underline{50}}$$

$$P(\text{Premium}) = 0.2$$

$$P(\text{Bau'e}) = 0.8$$

$$\rightarrow P(\text{Thriller} | \text{Premium}) = \boxed{0.8}$$

$$P(\text{Thriller} | \text{Basic}) = 0.5$$

$P(A|B)$ is known but they ask you to calculate $P(B|A)$ always apply Bayes' Theorem

$$P(\text{Premium} | \text{Thriller}) //$$

$$P(\text{Premium} | \text{Thriller}) = P(\text{Thriller} | \text{Premium}) \times P(\text{Premium})$$

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$$

$$P(\text{Thriller}) = P(\text{Thriller} | \text{Premium}) \times P(\text{Premium}) + P(\text{Thriller} | \text{Basic}) \times P(\text{Basic})$$

$$= 0.8 \times (0.2) + 0.5 \times 0.8$$

$$\begin{aligned}
 & \xrightarrow{0.56} \\
 & = \frac{P(\text{Thriller} | \text{Premium}) \times P(\text{Premium})}{P(\text{Thriller})} \\
 & = \frac{0.8 \times 0.2}{0.56} \\
 & = \frac{0.16}{0.56} = \underline{\underline{0.285}}
 \end{aligned}$$

Baye's Theorem

$$P(M) = 0.6 //$$

$$P(L) = 0.4 //$$

$$P(R|M) = 0.05 //$$

$$P(L|M) = 0.15 //$$

$$P(R|L) = 0.10 \Rightarrow P(L|R) = ? \quad \downarrow \quad \downarrow \quad = 4/7$$

$$P(L|R) = \frac{P(R|L) \times P(L)}{P(R)} = \frac{0.1 \times 0.4}{0.7}$$

$$\begin{aligned} P(R) &= P(R|M) \times P(M) + P(R|L) \times P(L) \\ &= 0.05 \times 0.6 + 0.10 \times 0.4 \\ &= 0.07 \end{aligned}$$

PMF, PDF, CDF

$$\begin{aligned} P(X=5) &\xrightarrow{\text{discrete}} \text{PMF} \\ P(X=4) &\rightarrow P(62.5 < X < 70) \rightarrow \text{PDF} \\ P(X < 180) &\rightarrow \text{CDF} \end{aligned}$$

$$P(\underline{62.5} < X < \underline{70})$$